

```

V H+ORROTATE X; INCL; I; OMEGA; O; OMEGA; Q
X+((pX),1)pX+ 1 0 1 \X
OMEGA+RADIAN PERIANGLE-ASCENDING
OMEGA+RADIAN ASCENDING
INCL+RADIAN INCLINATION
I+INCLROTATE INCL
O+LONGROTATE OMEGA
Q+LONGROTATE OMEGA
H+(Q TIMES I TIMES O) TIMES X
H+((1+pH),x/(1+pH)pH)

```

The rotations are achieved by a series of matrix products. The functions *INCLROTATE* and *LONGROTATE* generate the appropriate matrices of sines and cosines, stacking them in a three-dimensional array since several sets of coordinates are to be rotated at once. The function *TIMES* (not shown) calculates the ordinary matrix product of the corresponding pairs of matrices in a three-dimensional stack.

The functions *INCLROTATE* and *LONGROTATE* generate stacks of matrices containing the appropriate sines and cosines of the angles through which rotation is to occur (see Figure 6):

```

V Z+INCLROTATE INCL; RHO
RHO+P INCL
Z+((p,INCL), 3 3)p9+1
Z[;2;2]+Z[;3;3]+COS INCL
Z[;2;3]+Z[;3;2]+SIN INCL
+ (O<pRHO)/O
Z+ 3 3 pZ

```

```

V Z+LONGROTATE OMEGA; RHO
RHO+P OMEGA
Z+((p,OMEGA), 3 3)p 0 0 0 1 0 0 0
Z[;1;1]+Z[;3;3]+COS OMEGA
Z[;3;1]+Z[;1;3]+SIN OMEGA
+ (O<pRHO)/O
Z+ 3 3 pZ

```

These functions are used in translating the heliocentric coordinates of the planets to geocentric equatorial coordinates (i.e. the view from the center of the Earth):

```

V GQ+DATE EARTHVIEW H; GC
GC+H-(pH)pDATE PLANETPOS EARTH
GQ+3 RADECDIST GC+.xINCLROTATE -RADIAN AXITILT

```

in which *AXITILT* is the angle between the axis of the Earth and the plane of the ecliptic.

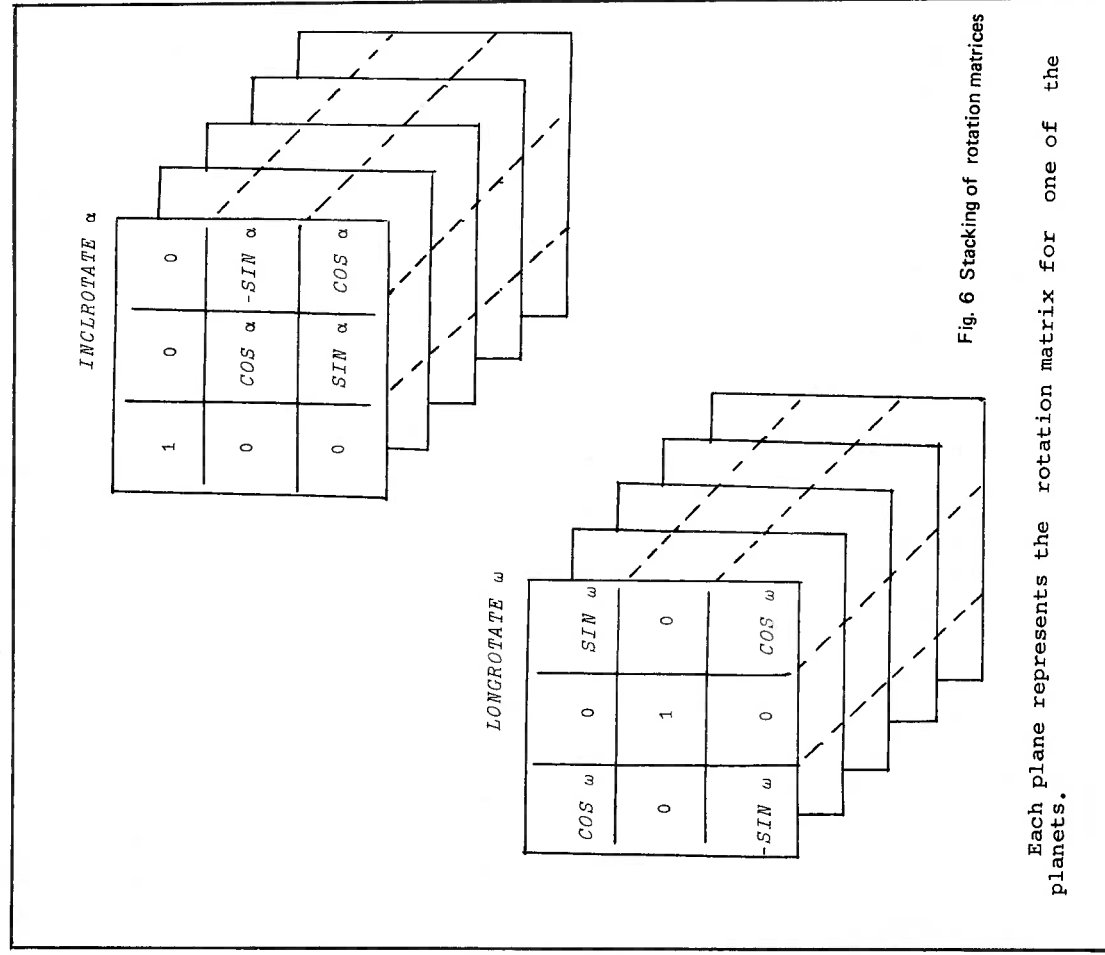


Fig. 6 Stacking of rotation matrices

Each plane represents the rotation matrix for one of the planets.